Which 16 of 24? Per-Die Silicon Harvest Maps, Procurement Lots and Core Guarding from Supercomputer Telemetry

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***Abstract*—Processor vendors sell partially defective dies by permanently fusing off the faulty units. The resulting per-die maps of *which* units were disabled are proprietary and conventionally invisible to the buyer. We show that the out-of-band sensor telemetry supercomputers already collect discloses them: a baseboard management controller exposes one thermal sensor per physical core position and does not renumber survivors, so the set of reporting sensors *is* the harvest map. We recover maps for all 1,962 POWER9 sockets of CINECA's Marconi100 from a public dataset [1] and track them across 1.67 million socket-days.**

Harvesting operates exactly at the POWER9 *slice*—the pair of cores sharing an L2/L3 block—with zero exceptions, and is per-die rather than per-SKU: 443 of the 495 possible patterns occur. A thermal probe over 240 sockets independently recovers the sensor index as the physical ordering of slices on the die ($z = -6.13$; an unconstrained search returns the same ordering reversed, $|\rho| = 0.993$), which establishes that harvested slices cluster *spatially*. A fitted pairwise-interaction model separates that clustering from per-slice propensity and shows a sharp procurement boundary at node 439: the two lots differ in both. Maps are static—changing only across reboots—and we isolate three field core-guard episodes. A taxonomy separates processor replacements from node replacements, socket relabelling and firmware deconfiguration, giving an annualised socket replacement rate of 1.96 % with no asset database. Finally, the harvest map has no measurable effect on socket power, so schedulers need not model it.

***Index Terms*—core harvesting, silicon binning, HPC telemetry, POWER9, yield, hardware provenance**

I. Introduction

Every large processor die ships with defects. Rather than discard them, vendors permanently fuse off the faulty units and sell the remainder as a lower-core-count part. We call this *silicon harvesting*, or *die harvesting*, to distinguish it from the now more common architectural sense of "core harvesting" as the reclamation of idle CPU *cycles* for co-located work [2]; the two are unrelated. The practice is universal and openly acknowledged, but the resulting *per-die* maps of which units were disabled are proprietary: a buyer is told the part has 16 cores, never *which* 16. Those maps encode information about a fab's defect process, a vendor's binning policy and a procurement's provenance that is otherwise unavailable outside the manufacturer.

Independently, HPC operators have begun publishing holistic out-of-band monitoring at fleet scale [1]. We observe that such telemetry carries an unintended side channel: a baseboard management controller exposes one thermal sensor per *physical* core position and does not renumber the surviving cores compactly, so the set of sensors that report *is* the die's harvest map, published inadvertently and at fleet scale.

We exploit this to recover harvest maps for every processor socket of CINECA's Marconi100, a Top-10 system, from a public dataset, and to track them across 858 days. Our contributions are:

- A recovery method (Sec. III) with an explicit validity argument, needing only per-core sensor telemetry that operators already collect.
- The first fleet-scale characterisation of core harvesting in a deployed supercomputer: the granularity is exactly the POWER9 slice, maps are per-die rather than per-SKU, and harvested slices cluster in sensor-index space.
- Evidence of *procurement-lot structure*—a sharp change-point separating two populations with markedly different binning statistics—from telemetry alone.
- A longitudinal result over 1.67 M socket-days: maps are static, configuration changes occur only at reboot, and three field core-guard episodes are isolated and corroborated.
- A taxonomy that separates processor replacements, node replacements, socket relabelling and firmware deconfiguration from one another (Sec. IV-I), yielding an annualised socket replacement rate of 1.96 % with no asset database.
- A negative result that matters operationally: the harvest map has no practical effect on socket power (Sec. IV-L),

so schedulers need not model it.

II. Background

A. The machine, from room to core

Marconi100's 980 compute nodes occupy 49 rack cabinets of 20 nodes each, arranged in three rows of the machine room, with $\mathrm{id} = 20\,\mathrm{rack} + \mathrm{slot}$ and every rack's (X, Y) floor position documented [3]. Each node is an AC922 8335-GTG chassis carrying *two* CPU sockets and four V100 GPUs. Each socket holds one POWER9 die; each die carries 24 SMT4 core positions organised as 12 slices; each slice is two cores plus their shared L2 and L3. Fig. 1 shows the chain. Two consequences matter throughout: the unit we recover a map for is a *socket*—two per node, not one—and the unit that gets fused is a *slice*, not a core. Sec. IV-A fixes the exact socket populations.

B. POWER9 slice organisation

The POWER9 die used in the IBM Power System AC922 contains 24 SMT4 cores (equivalently 12 SMT8 cores) on a $695\,\mathrm{mm}^2$ die in 14 nm FinFET with 17 metal layers [4]. Critically, *each pair of SMT4 cores—or one singleton SMT8 core—comprises a* slice*, and each slice owns 512 kB of L2 and 10 MB of L3* [4], [5]. IBM's Hot Chips 28 presentation shows the 24 cores above 12 × 10 MB L3 blocks, attached to a 7 TB/s on-chip fabric at $256\,\mathrm{GB/s} \times 12$ [5]. The slice is therefore the natural fusing granularity: disabling one SMT4 core would strand its half of a shared L2/L3 block.

Marconi100 nodes are AC922 model 8335-GTG with 2× 16-core POWER9 at 2.6/3.1 GHz and 4× NVIDIA V100 [6], [7]. **A 16-core part is a 24-core die with four slices fused off**; this is the object of our study. Fig. 2 shows the organisation annotated with the harvest rates we measure.

C. The M100 ExaData dataset

M100 ExaData [1] publishes 49.9 TB of telemetry from Marconi100 covering 2020-03 to 2022-09 across 980+ nodes. The IPMI plugin collects BMC sensor data at 20 s cadence. The documentation describes the core-temperature family as `pX_coreY_temp`, "Temperature of core n. Y in the CPU socket n. X. X=0..1, Y=0..23" [3]. *This specifies the BMC sensor-ID namespace, not the active core count*—the distinction is the entry point for this study.

III. Method

A. Recovering the disable map

For each month we enumerate the `metric=pX_coreY_temp` partitions and count samples per (node, socket, core). A core index is *present* for a socket if it emits any sample in the window. We define the disable map of socket s as $M[s, k] \in \{0, 1\}^{12}$ over slices $k = \lfloor \mathrm{core}/2 \rfloor$.

The key validity argument is that **the BMC does not renumber active cores compactly**: were that so, every socket would report indices 0–15. Instead we observe arbitrary 16-element subsets of $\{0, \ldots, 23\}$, so the index is a fixed physical-position identifier.

B. Pipeline, filters, and what each filter protects against

The recovery above is one line of logic; making it reliable at 1.67 M socket-days is mostly a question of filters, and we state them here rather than introducing them where they are first used.

Stage 1 — extraction. Each monthly archive is scanned once and only the 48 `p{0,1}_core{0..23}_temp` partitions are extracted. A **completeness gate** refuses to emit a month unless all 48 are present. This is not defensive boilerplate: a partial extraction is indistinguishable, downstream, from a socket with disabled cores, and an early version of this pipeline produced exactly that artefact for three months (Sec. III-C).

Stage 2 — daily aggregation. Samples are reduced to a count per (node, socket, core, day), giving a 15 MB table from a 400 GB scan. Every structural and longitudinal result in this paper is computed from that table; only the thermal analysis returns to the raw series.

Stage 3 — the clean-day filter. A socket-day is *clean* when exactly 16 cores report *and* each reports more than 90 % of the 4,320 samples expected at 20 s cadence. This is the load-bearing choice in the paper, so its effect is stated plainly: it retains 1,517,473 of 1,671,297 socket-days (90.8 %), and Sec. IV-G shows the lot result is unchanged when the threshold is tightened to sockets with $\geq$90 % clean-day coverage. The filter exists because a partially-collected day mimics a configuration change.

Stage 4 — per-socket map. A socket is summarised by the modal harvest map over its clean days (median 775 days supporting each map). Sockets whose modal map does not have exactly four harvested slices are excluded; this removes 6 of 1,968.

Cost. One full pass is a 400 GB sequential scan, roughly two hours on a single node, dominated by archive traversal rather than computation. Re-running every result from the derived table takes minutes. The asymmetry matters for reproducibility: we publish the derived table, so the expensive stage need not be repeated.

C. Failure modes of the method

Four things could make this recovery wrong, and each has a detector.

Partial ingestion. If archive extraction silently truncates, absent metrics look like harvested cores. *Detector:* the 48-metric completeness gate. We found this the hard way — three months were once written from incomplete extractions, one of them missing all 24 sensors of socket p1, which would have read as a socket with no active cores at all.

Compact renumbering. If a BMC renumbered surviving cores $0..15$, the method would recover nothing but a constant. *Detector:* we observe arbitrary 16-element subsets of $\{0, \ldots, 23\}$ and 443 distinct patterns; a renumbering BMC would produce exactly one.

Per-metric collector faults. A collector that drops individual metric streams mimics deconfiguration. *Detectors:* odd active-core counts, which slice-granular fusing cannot produce (39 socket-days); reduced counts that are strict subsets of the

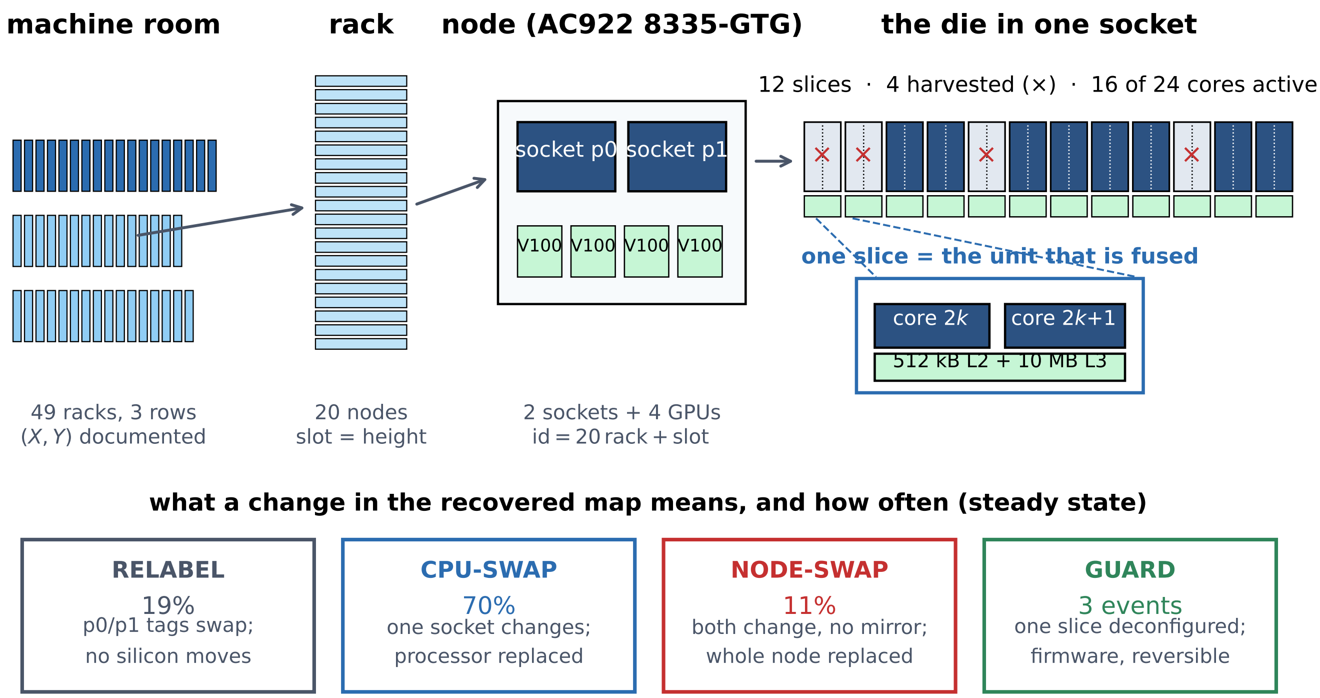

Fig. 1. The physical hierarchy this paper reasons over, and what a change in a recovered map means at each level. A socket's harvest map identifies the *die currently seated in that position*, so a change in the map is a change of die, of position labelling, or of configuration— and Sec. IV-I shows these are distinguishable from telemetry alone.

socket's modal map; and, for the guard episodes specifically, the control that every other sensor on the same socket kept full cadence (Sec. IV-H).

Socket-tag swaps. If the collector exchanges the p0/p1 tags, a longitudinal study silently mixes two dies. *Detector:* the mirror test of Sec. IV-I, which identifies 19 % of steady-state map changes as relabelling rather than replacement.

D. Null model for clustering

Every socket has exactly four harvested slices, so the constraint induces negative dependence and a naive independence null is invalid. We use the curveball algorithm [8], which preserves *both* row sums (4 per socket) and column sums (per-slice marginals) while randomising association structure. We use 200 draws of 20,000 swaps each, and assert row- and column-sum preservation on every draw. Repeating the whole null at 25–400 draws moves the resulting z by about ± 1, so we quote it to the nearest integer rather than to a decimal.

E. Thermal probing of die topology

Cores on a socket share workload and inlet air, producing a dominant common mode. We remove it by subtracting the per-timestamp cross-core mean, double-centre, and compute residual correlations at the sensors' native 20 s cadence—no resampling is applied, and Sec. IV-E shows why that matters. Physically abutted cores should retain positive residual coupling.

Caveat: Common-mode removal forces each row of the residual matrix to sum to approximately zero, so negative values at large index gaps are partly mechanical. We therefore rely on contrasts *at identical index distance*, which control for this exactly.

IV. Results

A. Every socket is 16 cores; granularity is the slice

Populations: Three socket populations recur and are kept distinct throughout: $\mathcal{S}_{\text{all}}$, the 1,968 sockets appearing anywhere in the record (984 nodes); $\mathcal{S}_{\text{map}}$, the 1,962 with at least one clean day and hence a well-defined harvest map; and $\mathcal{S}_{2004}$, the 1,960 present in the single month 2020-04. All structural results below are over $\mathcal{S}_{\text{map}}$ and the full 31-month record.

TABLE I
Disable-map structure over the full 31-month record.

Sockets with a well-defined harvest map	1,962
Median clean days supporting each map	775
Socket-days reporting exactly 16 cores	**99.985%**
Sockets ever reporting 24 cores	0
Harvested cores form complete $(2k, 2k{+}1)$ pairs	**1,962 (100%)**

The perfect pair structure is the central mechanical finding: it matches the POWER9 slice definition exactly and rules out per-core fusing.

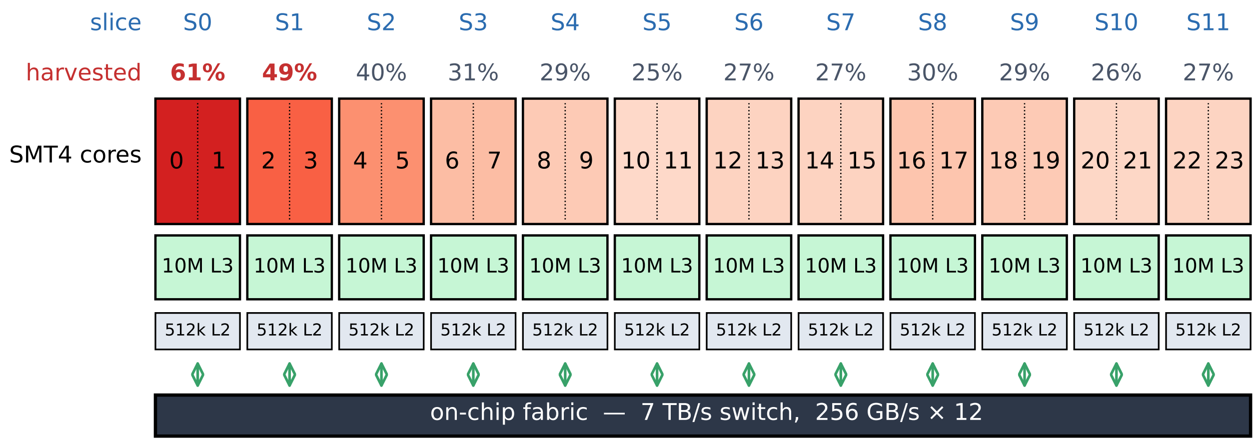

Fig. 2. POWER9 slice organisation (after [5]) annotated with the measured per-slice harvest rate across the 1,962 Marconi100 sockets with a well-defined map. Each slice holds two SMT4 cores sharing 512 kB L2 and 10 MB L3 on a common fabric port. Harvesting always removes whole slices. **The horizontal ordering is the IPMI sensor index, which Sec. IV-F shows is the physical die ordering, recovered independently from thermal coupling.**

B. The map is per-die, not per-SKU

443 distinct patterns appear among the $\binom{12}{4} = 495$ possible (Fig. 8). The most common (slices 0–3) covers only 5.5% of sockets. Within a node the two sockets match in just 1.2 % of cases (12 of 981 nodes), with mean overlap 1.641 slices against a shuffled null of 1.468 ± 0.026 ($z = +6.7$)—nearly independent, with a small excess attributable to same-lot pairing.

C. Marginals are strongly non-uniform

Slice 0 is harvested on 61.2% of sockets against a uniform expectation of 33.3% ($\chi^2 = 807.2$, df $= 11$, $p \ll 0.001$); Fig. 3 splits this by lot. These marginals are remarkably stable in time: across the 31 months the largest standard deviation of any slice's monthly rate is **0.25 percentage points**, and slice 0 ranges only 60.9–61.8% (Fig. 9). The single-month snapshot used in earlier drafts was therefore representative, and every estimate here is a 2.5-year-stable measurement.

D. Harvested slices cluster in index space

Under the curveball null the mean index gap between harvested slices is 4.242 versus 4.582 ± 0.017 ($z \approx -20$; Monte-Carlo noise ± 1, see Sec. III-D). Applying a Bonferroni correction across all $\binom{12}{2} = 66$ pair tests ($|z| > 3.83$), 17 pairs remain significant (Fig. 4). The positive excesses are 0&1 ($z = +16.7$), 6&7 (+9.2), 2&3 (+8.0), 8&9 (+7.7), 10&11 (+7.5), 4&5 (+7.1), 1&2 (+6.9) and 0&2 (+6.3): **eight of the nine positive-excess pairs have** $|\Delta k| \leq 2$, and every significant *deficit* is a distant pair. We report the single exception rather than rounding it away. Because Sec. IV-F shows index adjacency *is* physical adjacency, this is spatial clustering, not merely index-space clustering.

E. Slice siblings are physically abutted

We probe 240 sockets—120 nodes, *both* p0 and p1—across the full 31-day month 2022-08 (Fig. 5). At *identical* index distance of 1, within-slice siblings are more strongly coupled than cross-slice neighbours, and the ordering is unambiguous: at native 20 s resolution $r = +0.377$ (95 % CI $[0.366, 0.387]$) against $+0.308$ ($[0.291, 0.324]$), a **contrast of** $+0.069$ with permutation $p < 10^{-5}$ and non-overlapping intervals. The effect replicates independently on each socket. The two cores of a slice are measurably more coupled than equally-numbered cores across a slice boundary, confirming physical abutment and the slice as a physical unit.

The effect size depends on temporal resolution; the conclusion does not: Cores are sampled every 20 s and their timestamps coincide across a socket, so no resampling is required—an inner join at native resolution retains 133,907 of 133,920 possible rows (99.99 %). Earlier versions of this analysis nonetheless aligned onto a one-minute grid, a step introduced to work around an unrelated bug and never revisited. It is not neutral (Table III): coarser averaging suppresses high-frequency noise and inflates every correlation, within-slice more than cross-slice, so within this subsample the contrast grows from $+0.063$ at 20 s to $+0.140$ at 15 min. **We therefore quote the native-resolution figure, which is the most conservative.** What does *not* move is the ordering—within-slice exceeds cross-slice at every grid—and, critically, the floorplan fit of Sec. IV-F, which sits at $-0.787, -0.773, -0.766, -0.767$ across the four grids. The paper's load-bearing thermal claim is independent of this choice.

Abutment or shared cache?: The two cores of a slice are physically adjacent *and* share an L2/L3 block, so shared *activity* is an alternative explanation for the excess that would need no thermal path. Core-to-core thermal time constants on a die are far shorter than a 20 s sample, so both mechanisms peak at lag zero; what distinguishes them is the *shape* of the lagged profile. Instantaneous activity sharing predicts a sharp lag-0 peak decaying with the workload's own autocorrelation. We measure the opposite (Table II): the excess is *smallest* at lag 0 and grows monotonically to $1.66\times$ that value at ± 10 samples (200 s), while the cores' own autocorrelation falls to 0.60 over the same span. A broad, lag-persistent excess is the signature of a slowly varying shared physical environment. (Both curves spike at lag 0 relative to their neighbours, consistent with a shared same-sample measurement artefact—so the excess is least trustworthy at exactly the lag an activity explanation would need it to be largest.)

TABLE II
LAGGED CROSS-CORRELATION, 60 NODES, COMMON MODE REMOVED, NATIVE CADENCE. THE WITHIN-SLICE EXCESS GROWS WITH LAG; INSTANTANEOUS ACTIVITY SHARING PREDICTS THE OPPOSITE.

lag (20 s)	within	cross	excess	core autocorr.
0	0.3522	0.3023	0.0499	1.000
± 1	0.2759	0.2144	0.0615	0.680
± 2	0.2804	0.2141	0.0663	0.663
± 5	0.2858	0.2110	0.0748	0.630
± 10	0.2907	0.2077	0.0830	0.604

TABLE III
THERMAL RESULTS AGAINST ALIGNMENT GRID. THIS SWEEP USES A 180-SOCKET SUBSAMPLE, SO ITS 20 S ROW DIFFERS MARGINALLY FROM THE 240-SOCKET PRIMARY ESTIMATE ABOVE; THE POINT IS THE TREND ACROSS ROWS, NOT THE ABSOLUTE VALUES.

Grid	rows/socket	within	cross	1×12 fit
20 s (native)	**133,907**	**+0.377**	**+0.314**	-0.787
1 min	44,640	+0.411	+0.329	-0.773
5 min	8,928	+0.466	+0.351	-0.766
15 min	2,976	+0.508	+0.368	-0.767

A correction to an earlier estimate: An initial version of this analysis used 40 sockets in 2020-04, a month with only three reporting days, and reported a contrast of $+0.51$ vs $+0.29$. That overstated the effect roughly threefold. The direction, the significance and the replication across sockets survive; the magnitude does not. We report the full-month estimate.

F. The sensor index recovers the physical slice ordering

Is the sensor index merely a label, or does it track physical position? We answer it by scoring explicit candidate floorplans: for each, correlate the predicted physical distance between two slices against their measured thermal coupling (Fig. 6). Coupling should fall with distance, so a better floorplan gives a more negative correlation.

The identity 1×12 linear arrangement wins decisively, at -0.754 against a random-ordering null of $+0.001 \pm 0.123$ ($z = -6.13$, $p < 10^{-4}$), and beats every two-dimensional alternative tried (6×2: -0.716; 4×3: -0.555; 3×4: -0.488; 2×6: -0.213). More stringently, an unconstrained hill-climbing search over one-dimensional orderings—free to return any of the $12!/2$ arrangements—converges to $[10, 11, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0]$ at -0.759, barely better than the identity's -0.754. That is the sensor ordering *reversed*; since distance is invariant under reflection it is the same physical arrangement, differing by one adjacent transposition. Rank correlation with the identity is $|\rho| = 0.993$.

This result cannot be cache sharing: The slice-level coupling above is computed from **cross-slice pairs only**, and two cores in different slices share no L2 and no L3. Measured over those pairs alone, coupling decays monotonically with slice distance—from $+0.215$ at adjacent slices to -0.353 at distance 10, giving corr(distance, coupling) $= -0.929$. Whatever explains the within-slice excess, it cannot explain this: the pairs that share cache are excluded from the floorplan estimate by construction.

The thermal data therefore recovers the sensor-index ordering independently, up to reflection. This licenses reading Sec. IV-D spatially: slices adjacent in index are adjacent on the die, so the co-harvest excess is genuine *spatial* clustering of the kind the yield literature describes [9], [10].

A correction to an earlier estimate: On the three-day 2020-04 sample this test was inconclusive—the linear ordering scored only -0.283 ($p = 0.023$) and was beaten by a 3×4 grid—and we previously declined to claim a physical ordering. The full-month, both-socket measurement reverses that conclusion. We report both so the reader can judge how much the earlier sample size mattered.

G. A procurement-lot boundary at node 439

What a rack is here: Marconi100's nodes occupy 49 physical rack cabinets of 20 nodes each, with $\mathrm{id} = 20\,\mathrm{rack} + \mathrm{slot}$, and the dataset documents every rack's (X, Y) position on the machine-room floor in three rows [3]. The rack index is a real physical coordinate, not a label—which is why it is informative, and why it must be handled carefully: it conflates *procurement batch* (hardware arrives and is racked in deliveries, in id order) with *room position* (cooling, airflow, height in cabinet). We separate the two below.

A changepoint scan maximises Welch t at rack 22, $t = 26.6$ (Fig. 7). At node granularity the boundary is sharper still: $t = 26.8$ between nodes 438 and 439, with the 40 sockets immediately before at 95.0 % slice-0 harvested and the 40 immediately after at 42.5 %. Node 439 sits in the topmost slot of rack 21 yet behaves like the later population (0 of its 2 sockets have slice 0 harvested, against 2 of 2 for each of nodes 430–438), so the delivery boundary is rack-aligned to within a single node.

TABLE IV
TWO POPULATIONS EITHER SIDE OF THE LOT BOUNDARY.

Population	Sockets	P(slice 0)	Distinct patterns
Lot A (racks 0–21)	880	88.2%	230
Lot B (racks 22–48)	1,082	39.3%	416

Lot A marginals fall steeply with slice index; Lot B is nearly flat. Permutation tests confirm between-rack heterogeneity for slices 0, 1, 2, 5 ($p < 0.01$), and same-rack sockets share more harvested slices than chance (1.565 vs 1.452 ± 0.008, $z = +13.9$).

Two deliveries, not more: Recursive binary segmentation over the node-ordered sequence, splitting wherever $|t| > 4$, finds *no* changepoint beyond the primary one. The fleet is two populations—nodes 0–438 at 88.4 % and 439–979 at 39.1 %—not a gradient and not a sequence of many small batches.

Procurement, not room position: Because rack index conflates batch with geometry, we test the geometric alternative directly. Raw correlations with slice-0 status are substantial for the machine-room row ($r = -0.414$ for Y) and visible for aisle position ($r = +0.124$), exactly as expected when both track rack index. **Once the lot is accounted for, they vanish**: partial correlations are $+0.017$ for Y ($p = 0.44$) and $+0.014$ for X ($p = 0.53$). Only the slot—height within the cabinet—retains a residual -0.054 ($p = 0.017$, 0.3 % of variance), which is marginal under a three-test correction and would in any case be consistent with racks being filled in delivery order.

This is the empirical form of an argument that also holds *a priori*: fusing happens at manufacture, months before a part is installed, so no property of the machine room can influence a harvest map. The only admissible causal direction runs from procurement batch to rack position.

The boundary is not an artefact of the coverage filter: Because every structural result is computed on *clean* socket-days, a lot-correlated dropout could in principle manufacture this contrast. It does not. Clean-day rates are near-identical between lots (0.909 vs 0.902), coverage is uncorrelated with the harvest pattern (corr with slice-0 status $= +0.004$, $p = 0.88$), and restricting to the 1,847 sockets with $\geq$90 % clean-day coverage leaves the contrast unchanged at $+49.0$ percentage points versus $+48.9$ over all sockets.

H. Configuration lifetime

POWER9 exposes mechanisms that *could* alter the active set after manufacture: predictive Dynamic Processor Deallocation, persistent GARD records read by hostboot at IPL and clearable with `opal-gard`, and the field core override used for licensing [11], [12]. Operating-system core offlining is invisible to the BMC and cannot affect sensor presence. Whether harvest maps are static in practice is therefore an empirical question, which we answer by aggregating every core sample to (node, socket, day) counts across the dataset. All numbers in this section are over the complete 31-month record (2020-03 to 2022-09).

a) The count is very nearly invariant: Across all 31 months—1,671,297 socket-days over 858 days and 984 nodes—**99.985% report exactly 16 active cores**. Of the 256 exceptions, 39 have an odd count, which *cannot* arise from slice-granular fusing and therefore proves the collector can drop individual cores; the counts above 16 are unions spanning a same-day reconfiguration. Only **nine** socket-days are fully-sampled with an even count below 16, and these are the subject of the guard analysis below.

b) Configurations are near-static: Restricting to *clean* socket-days (16 cores, each at >90% of the 4,320 samples expected at 20 s cadence) leaves 1,517,473 socket-days, a median of 775 observed days per socket. **89.5% of sockets never change configuration** (1,756 of 1,962); 183 have two distinct sets, 19 have three, three have four and one has five.

c) Three core-guard episodes: The nine anomalous socket-days resolve into **three sustained episodes on three distinct sockets**, each with the exact signature of a field deconfiguration (Table V). In every case the socket loses *exactly one slice*—a complete $(2k, 2k{+}1)$ pair—runs on 14 cores for 2–4 consecutive fully-sampled days, and in at least one case the slice is subsequently restored, consistent with an operator clearing the GARD record [12]. The measured rate is **3 episodes across 1,968 sockets over 2.5 years**, i.e. 0.15% of sockets. Durations are lower bounds, since an episode is counted only on fully-sampled days.

Ruling out a collector fault. The competing explanation is that the collector dropped exactly those two metric streams. We test it directly: for each episode we ask whether the *other* sensors on the same node—in particular on the same socket—kept reporting. They did. In all three episodes the two slice sensors read zero for the entire window while **all ten control metrics, including `pX_power` and `pX_vdd_temp` on the same socket, sustained a full 4,320 samples per day** (Fig. 10). A collection fault that silences precisely the two cores of one slice for several consecutive days, while leaving every other sensor on the same socket at full cadence, is not a plausible account. We therefore attribute these episodes to hardware deconfiguration; to our knowledge this is the first direct observation of POWER9 field core guarding in a deployed system.

TABLE V
CORE-GUARD EPISODES. BASELINES ARE THE ADJACENT OBSERVED DAY, NOT THE SOCKET'S GLOBAL MODAL SET, SINCE TWO OF THESE SOCKETS ALSO CHANGE CONFIGURATION ELSEWHERE IN THEIR LIFE.

Socket	Dates	Days	Slice lost	Restored
347 p1	2020-03-13–16	4	4 (cores 8,9)	yes
411 p1	2021-11-13–15	3	6 (cores 12,13)	yes†
946 p0	2020-08-31–09-01	2	8 (cores 16,17)	not observed

†restored via a day that also spans a reconfiguration.

d) Changes never occur during operation: Of **235** transitions across the full record, **zero** occur between consecutive reporting days: every one coincides with a gap in reporting (median 2 d), against a background in which only 4.2% of consecutive reporting days are more than one day apart. Configuration changes are boot-time events, exactly as the guard/deconfiguration mechanism predicts.

The only two apparent counterexamples confirm the rule once examined at full timestamp resolution. Nodes 315 (p1) and 342 (p0) appear to change within a single day, and produce the dataset's only 20- and 22-core socket-days. In fact both cut over on 2021-03-02 across a *sub-day reporting gap*: on node 315 the outgoing cores $\{4, 5, 20, 21\}$ report until 08:42 and the incoming cores $\{8, 9, 10, 11\}$ resume at 09:10, a 28-minute outage, with exactly 16 cores before and 16 after; the 20/22-core counts are unions across the cutover. Node 342 cuts over in the same window, indicating a coordinated maintenance reboot. Day-granularity aggregation is therefore *conservative* here: it cannot see sub-day downtime, so our transition-gap statistics understate how consistently changes coincide with reboots.

e) Changes concentrate at commissioning: The 235 transitions involve 132 nodes on 81 distinct dates, but are heavily clustered: 2020-03-20 alone accounts for 63 and the top three dates for 41 %. Such synchronised, fleet-wide changes are characteristic of a newly commissioned machine—hardware is still being reconfigured and the monitoring pipeline is not yet settled—rather than of steady-state operation. The next paragraph shows this directly and excludes the period.

f) Commissioning period, identified and excluded: Two independent signals locate the end of commissioning without assuming a date. Telemetry coverage—the fraction of socket-days that are clean—is 0.82, 0.00 and 0.40 in 2020-03, -04 and -05, then rises to 0.72–1.00 for every one of the following 28 months. The transition rate falls from 91 in 2020-03 to a median of 4 per month thereafter. We therefore treat 2020-03 to 2020-05 as commissioning and re-run every result on 2020-06 onward:

- **The harvest maps are unchanged**: for all 1,960 sockets with a map in both windows, the recovered map is *byte-identical* (100.0 %). Marginals agree to 0.1 percentage points and the pattern count is 443 either way.
- **The lot boundary is unchanged**: Lot A 88.2 %, Lot B 39.2 %, Welch $t = 26.6$.
- **Zero transitions during operation still holds**: of the 136 steady-state transitions, *none* occurs between consecutive reporting days.
- **The fleet-wide clustering disappears with the commissioning window**: the largest single-date cluster in steady state is 11 events (8 % of steady-state transitions), against 63 (27 %) on 2020-03-20.

Every structural and longitudinal claim in this paper therefore rests on the post-commissioning period, and none of them depends on the anomalous early months.

g) Some apparent changes are socket relabelling: When node n's socket p0 loses exactly the slices p1 gains and vice versa, the dies are unchanged and only the p0/p1 labels have swapped. This accounts for **20%** of the 230 transitions where both sockets are observable.

h) Caveat: An apparent change can arise from (a) genuine hardware reconfiguration, (b) socket relabelling, or (c) a change in the BMC sensor-ID mapping, e.g. after a firmware update. Telemetry alone cannot separate these. The tight clustering of 86 transitions into three days of March 2020 is consistent with (c)—but that window is precisely the commissioning period excluded above, and every result is reported on the steady-state period where no such clustering exists. The ambiguity is therefore bounded rather than load-bearing.

I. A taxonomy of hardware changes

A harvest map identifies the die in a socket, so a change in the map is informative about what physically happened—but only if the possible causes can be told apart. They can

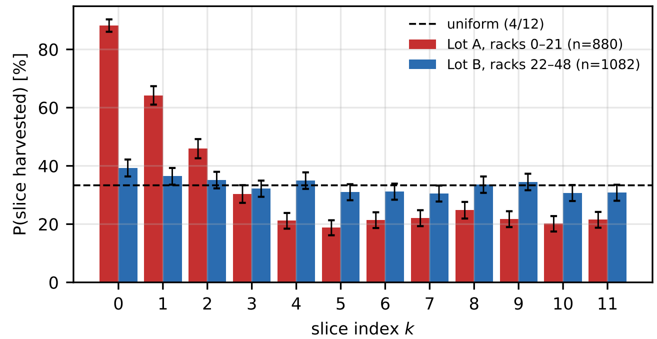

Fig. 3. Per-slice harvest probability by procurement lot.

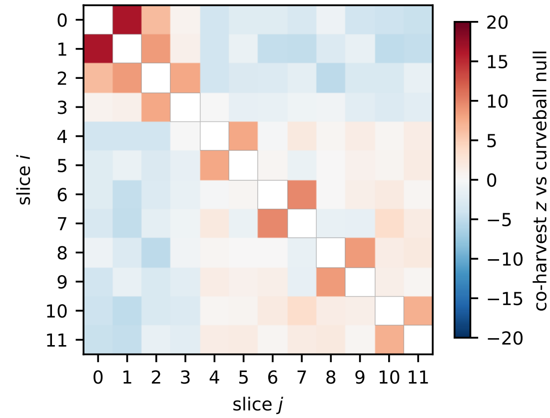

Fig. 4. Co-harvest z against the curveball null. Excess is confined to near-diagonal (index-adjacent) slice pairs.

(Fig. 1, bottom). If p0 loses exactly the slices p1 gains and vice versa, the same two dies are present and only the tags moved (RELABEL). If exactly one socket's map changes while the other is observed and unchanged across the same interval, one processor was replaced (CPU-SWAP). If both change and it is not a mirror, the node or both processors were replaced (NODE-SWAP). And a count dropping below 16 by exactly one slice, later restored, is a firmware deconfiguration with no hardware movement at all (GUARD, Sec. IV-H).

TABLE VI
HARDWARE-CHANGE EVENTS, CLASSIFIED AT NODE LEVEL.

Kind	Commissioning	Steady state	Interpretation
RELABEL	3	20	no silicon moved
CPU-SWAP	11	68	one processor replaced
CPU-SWAP?	2	6	as above, partner unobserved
NODE-SWAP	40	11	node or both processors replaced
Total	56	105	

The split validates the taxonomy on its own terms. During commissioning NODE-SWAP dominates (40 of 56, 71 %)—whole machines are being installed and reconfigured. In steady state the picture inverts: CPU-SWAP accounts for 70 % and

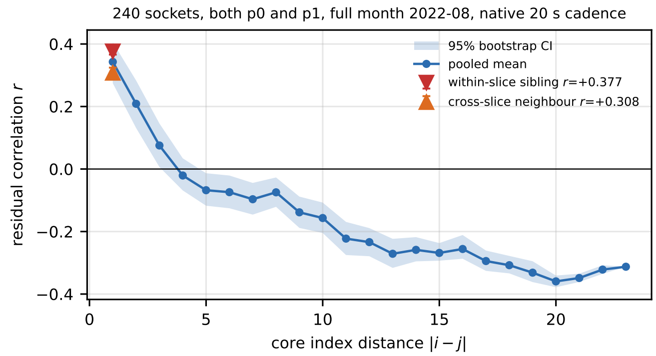

Fig. 5. Residual thermal correlation versus core index distance. The within- versus cross-slice contrast at distance 1 controls exactly for index gap.

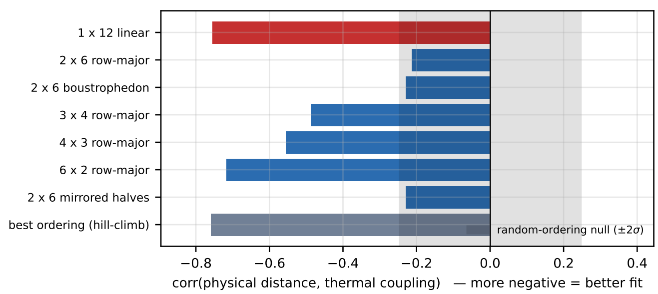

Fig. 6. Candidate floorplans scored against measured thermal coupling (more negative = better fit). The linear index ordering (red) wins decisively and is matched only by an unconstrained search that converges to the same arrangement reversed.

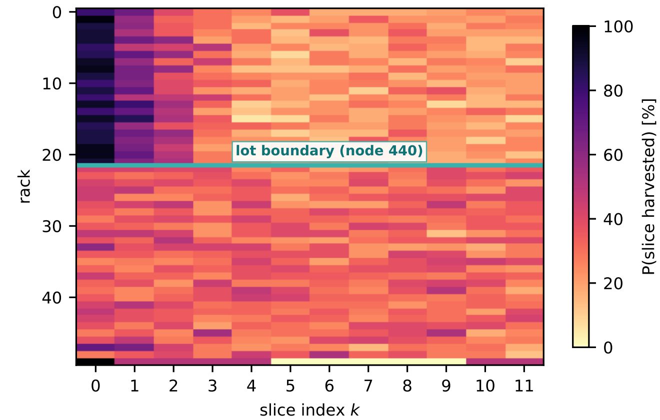

Fig. 7. Harvest probability by rack (rows: the 49 physical cabinets, 20 nodes each, in installation-id order) and slice (columns: position on the die). The procurement boundary is visible as an abrupt change across the whole die at once—the signature of a different population of parts, not of a room effect.

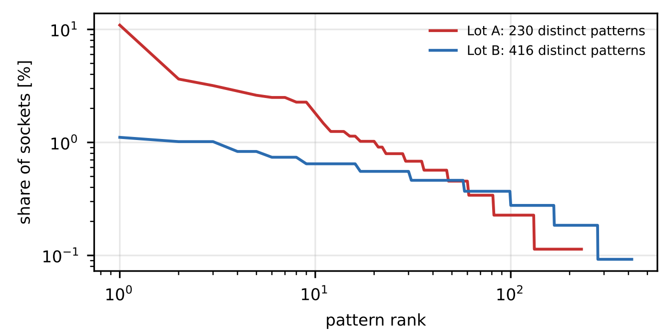

Fig. 8. Rank–frequency of disable patterns. Lot B is markedly more diverse.

NODE-SWAP for 11 %, which is what a maintenance regime that replaces individual failed processors looks like. Steady state contains 105 events on 87 distinct nodes over 28 months, about 3.8 per month.

Two consequences are worth stating. First, **19 % of steady-state map changes move no silicon**; a study that read every change as a replacement would overcount by that much. Second, counting one die per CPU-SWAP and two per NODE-SWAP gives **96 processor replacements among 1,960 sockets over 2.5 years—an annualised socket replacement rate of 1.96 %**, recovered without any asset database or maintenance log. Replacements are mildly clustered by rack (37 of 49 racks touched; concentration $z = +2.55$), consistent with correlated failures or with servicing being batched by cabinet.

J. A clustering parameter, fitted per lot

Section IV-D establishes clustering against a null; it does not *quantify* it, and the yield models we invoke [9], [10] are parameterised by a clustering strength. Those models describe defect *counts* per die, which we cannot observe: the SKU fixes the count at exactly four harvested slices. What is observable is the conditional spatial arrangement given four, so we fit the natural conditional analogue—a pairwise-interaction model on the twelve slice positions,

$$P(S) \propto \exp\Big(\textstyle\sum_{k\in S}\beta_k + \gamma\,\mathrm{adj}(S)\Big),$$

where $\mathrm{adj}(S)$ counts index-adjacent pairs in the harvested set S. The β_k absorb per-slice propensity (the slice-0 asymmetry) and γ measures clustering *beyond* what those propensities explain. The normalising constant is an exact sum over all $\binom{12}{4} = 495$ patterns, so likelihood and gradient are exact.

TABLE VII
FITTED CLUSTERING PARAMETER BY PROCUREMENT LOT. UNCERTAINTIES ARE **RACK-CLUSTERED BOOTSTRAP** STANDARD ERRORS (400 RESAMPLES OF THE 49 RACKS), NOT THE I.I.D. VALUES, BECAUSE SOCKETS ARE NOT INDEPENDENT—SEE THE TEXT.

Population	n	γ	LR χ^2_1 vs γ=0
Lot A (racks 0–21)	880	$+0.743 \pm 0.046$	174.2
Lot B (racks 22–48)	1,082	$+0.434 \pm 0.055$	106.4
Whole fleet	1,962	$+0.647 \pm 0.046$	394.5

Three things follow. First, **clustering is real in both lots** (χ^2_1 of 174 and 106, $p \ll 0.001$): mean adjacent-pair counts are 1.463 and 1.218 observed against 1.203 and 0.993 under the no-clustering model. Second, **the lots differ significantly in clustering strength**, not only in their marginals. Here the uncertainty needs care: an i.i.d. likelihood over sockets would give $z = +5.22$, but sockets are *not* independent—each node contributes two, and Sec. IV-G shows same-rack sockets share more harvested slices than chance. Resampling *racks* rather than sockets inflates the standard errors by $1.05\times$ to $1.58\times$ and moves the difference to $\gamma_A - \gamma_B = +0.309$, $z = +4.34$, 95 % CI $[+0.183, +0.456]$. The claim survives the correct uncertainty; we report the clustered figure. Third, and most usefully, the two effects *separate*: Lot A's $\beta_0 = +2.51$ dwarfs every other propensity while Lot B's β vector is nearly flat (range $+0.34$ to -0.13), yet both carry a substantial positive γ. The slice-0 asymmetry and the spatial clustering are therefore distinct phenomena rather than two views of one.

K. Sensitivity to the analysis choices

Every number above depends on three discretionary choices: where to set the clean-day threshold, how to summarise a socket's clean days into one map, and how many draws to run the null for. We varied all three (Table VIII).

TABLE VIII
SENSITIVITY OF THE HEADLINE RESULTS TO THE ANALYSIS CHOICES. THE PAPER'S SETTINGS ARE IN BOLD.

Choice	Clean s-days	Sockets	Patterns	$P(s_0)$	Welch t
clean-day threshold (fraction of the 4,320 expected samples)					
0.50	1,607,604	1,962	442	61.2 %	26.6
0.80	1,550,864	1,962	442	61.2 %	26.6
0.90	**1,517,473**	**1,962**	**443**	**61.2 %**	**26.6**
0.95	1,474,479	1,962	443	61.2 %	26.6
0.99	1,324,982	1,960	443	61.2 %	26.6
map rule (threshold 0.90); "differs" = sockets whose map changes					
mode	—	**1,962**	**443**	**61.2 %**	**26.6**
first clean day	—	1,962	442	61.6 %	28.4
last clean day	—	1,962	443	61.1 %	25.9
strict (all days equal)	—	1,756	436	60.8 %	28.0

The clean-day threshold is inert. Moving it from 0.50 to 0.99 discards 18 % of socket-days and changes nothing that matters: the same 1,962 sockets (1,960 at 0.99), 442–443 patterns, an identical 61.2 % slice-0 rate and an identical lot contrast. This is a stronger statement than "we chose 0.90".

The map rule matters slightly, and in the expected direction. Taking the first clean day instead of the mode changes 163 of 1,962 maps (8.3 %) and taking the last changes 49 (2.5 %) — these are exactly the sockets that undergo a transition, and the asymmetry (more change under "first" than "last") reflects that commissioning-era configurations are the ones later replaced. The lot contrast ranges only over t = 25.9–28.4. The *strict* rule, which keeps only sockets whose every clean day is identical, retains 1,756 sockets — precisely the 89.5 % that never change — and still reproduces the boundary at $t = 28.0$.

The null is converged. Running the curveball null at 25, 50, 100, 200 and 400 draws gives $z = -18.3, -18.6, -17.9, -19.6, -20.0$: stable in magnitude with about ± 1 of Monte-Carlo noise, which is why Sec. IV-D quotes $z \approx -20$.

The thermal alignment grid was the one remaining untested choice; it has since been varied over 20 s–15 min and is reported in Sec. IV-E. It is the only choice we found that materially moves an effect size, which is why the affected number is now quoted at native resolution.

L. The harvest map has no operational consequence

Recovering a hidden hardware property is only useful if it changes something. We therefore ask whether the harvest map predicts socket behaviour, using a **within-node paired contrast**: the two sockets of a node share chassis, airflow, inlet air, power supply and—to first order—workload, so the difference $p_0 - p_1$ removes the node-level confounds that make cross-node power comparisons unreliable. We summarise each socket's map by the mean index of its active slices, their dispersion, the number of adjacent active pairs, and the count of active slices in the low half, and correlate the socket-to-socket difference in each against the socket-to-socket difference in mean CPU power over six months (5,880 node-months, Fig. 12).

The associations are statistically detectable and practically negligible. The strongest is between the mean active-slice index and socket power: $r = +0.039$ (permutation $p = 0.003$), explaining **0.16% of variance**, with a slope of $+0.46$ W per unit of slice index (95% CI $[+0.16, +0.76]$). Across the full observed range of the predictor this is **3.1 W**, against a socket-to-socket power spread of 17.8 W. For calibration, the same paired design shows socket p0 draws 7.7 W more than socket p1 on average ($t = 44.6$)—mere physical position in the chassis matters more than twice as much as the entire range of harvest-map variation.

We therefore report a deliberate negative result: **harvest maps are operationally invisible**. Schedulers, power models and node-ranking heuristics need not represent them, and node-to-node performance variability in this system must be sought elsewhere. The result also strengthens the provenance interpretation of Sec. IV-G: because the map has no behavioural footprint, the lot boundary cannot be an artefact of operational differences between racks.

V. Discussion

a) Why is slice 0 special?: The single most striking number is that slice 0 is harvested on 88.2% of Lot A sockets but only 39.3% of Lot B. Two explanations are live. Under *defect-driven* harvesting, fusing follows wherever killer defects land. Under *policy-driven* harvesting, slice 0 abuts a shared structure and is preferentially fused for reasons of yield policy, binning recipe or physical design; an 88% rate on one specific slice is very hard to obtain from random defects.

The fit of Sec. IV-J shows the answer is *both*, and separates them. The propensity vector β carries the slice-0 asymmetry and differs enormously between lots ($\beta_0 = +2.51$ versus a flat Lot B); the interaction parameter γ carries the spatial clustering and is significantly positive in *both* lots. A defect process is operating throughout—consistent with the yield literature [9], [10]—while a strong additional slice-0 preference operates in Lot A only. The most economical reading is a change of binning recipe or die source between the two procurements, superimposed on a defect process present in both. We cannot settle the mechanism from one machine, but the decomposition is sharper than an either/or.

b) The direction of causation is fixed.: A tempting objection to Sec. IV-G is that rack position is confounded with cooling and workload. It cannot be: harvesting is performed by fusing at manufacture, months before a part is installed, so nothing about a socket's position in the machine room can influence its harvest map. The only admissible causal direction runs from procurement batch to rack, which is precisely the provenance signal we claim.

c) What this buys an operator.: Three things. Fleet telemetry can audit a procurement without vendor cooperation, detecting whether a delivery is homogeneous. It can detect field deconfiguration events that would otherwise be visible only in service logs. And, per Sec. IV-L, it establishes when a hardware property can safely be *ignored*.

VI. Related Work

All references in this section were verified against their original sources on 2026-08-05. Manufacturing variability in HPC has been studied mainly through its *performance* consequences—frequency and power spread under a power cap [13], turbo-boost variation across nominally identical processors [14], and accelerator-level variability in GPU-rich systems [15]. That literature treats the die as a black box characterised by its behaviour. Yield modelling, conversely, treats defect statistics directly: the negative-binomial family [9] and spatial wafer-map models [10] describe clustered killer defects but are validated on fab data unavailable to system owners.

A third, closer body of work concerns hardware provenance. Physical unclonable functions and inherent hardware identifiers exploit analog manufacturing variation to answer *which die is this*, for traceability and counterfeit detection [16]. That is a different question: we do not identify a die, we recover *what the vendor disabled on it*, without designed-in circuitry or physical access. Node-level anomaly and outlier detection in HPC [17], [18] likewise establishes *that* nodes differ without recovering the hardware structure responsible. The M100 ExaData release itself has an active downstream literature spanning telemetry foundation models, energy analytics and operational-data ontologies [19], [20], [21]; none of it examines per-core sensor presence.

We are not aware of prior work recovering per-die harvest maps from deployed-system telemetry, and that is the gap this paper addresses. We state this as an awareness claim rather than a priority claim: it rests on a systematic but necessarily incomplete search.

VII. Threats to Validity and Open Issues

Every unresolved item is tracked with an identifier in `OPEN_ISSUES.md` in the artefact repository, which records what is known, what would settle it, and where to look. **The items marked *blocking* below must be resolved before this paper is submitted.**

1) **Sensor presence $\neq$ core existence.** Mitigated by 99.985% of 1.67 M socket-days reporting exactly 16 cores, agreement with the published 16-core SKU [6], the perfect slice-pair structure, and the control of Sec. IV-H showing that when core sensors do vanish, every other sensor on the socket keeps reporting.
2) **[OI-1] Resolved.** The sensor index is recovered as the physical slice ordering from thermal coupling (Sec. IV-F), so the spatial reading of Sec. IV-D is supported. It remains an inference from thermal coupling rather than from a vendor floorplan; an annotated die shot would confirm it directly.
3) **[OI-2] Bounded.** A March 2020 BMC firmware update could have remapped sensor IDs, mimicking reconfiguration; 86 of the 235 transitions fall in the three days 2020-03-19–21. Those days lie inside the commissioning window excluded in Sec. IV-H0f, and every result is reported on the steady-state period, where the harvest maps are byte-identical and no comparable clustering

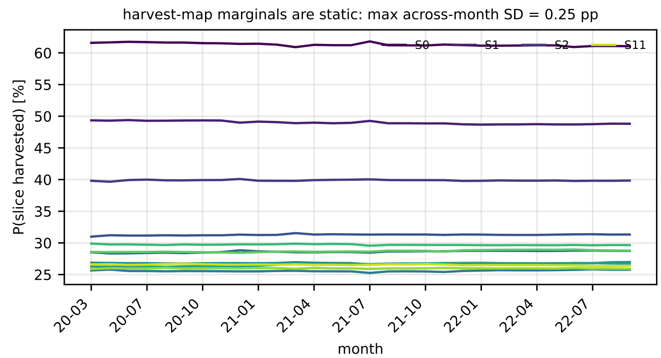

Fig. 9. Per-slice harvest rate in each of the 31 months. The largest across-month standard deviation of any slice is 0.25 percentage points.

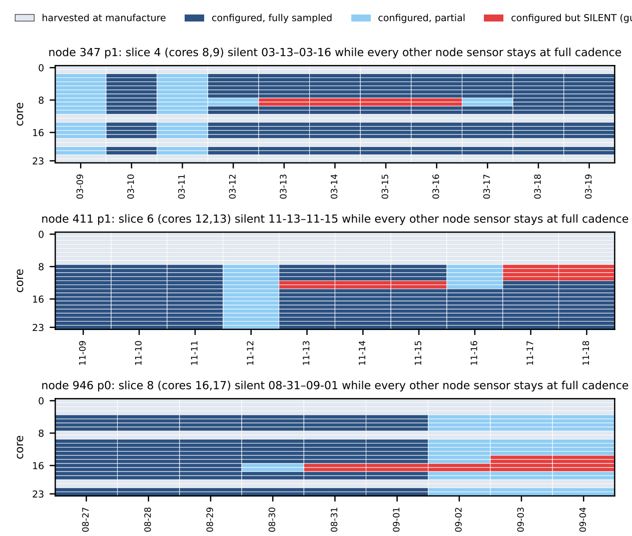

Fig. 10. The three core-guard episodes. Grey = harvested at manufacture, blue = configured and reporting, red = configured but silent. In every case exactly one slice goes silent while all ten control sensors on the same node stay at full cadence.

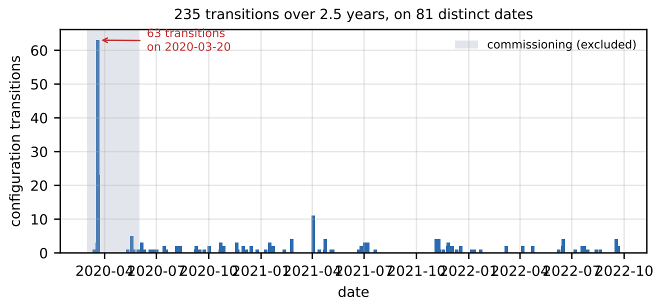

Fig. 11. Configuration transitions over the full record. 2020-03-20 alone accounts for 63, all inside the commissioning window (shaded) that Sec. IV-H0f excludes; the steady-state period shows no comparable cluster.

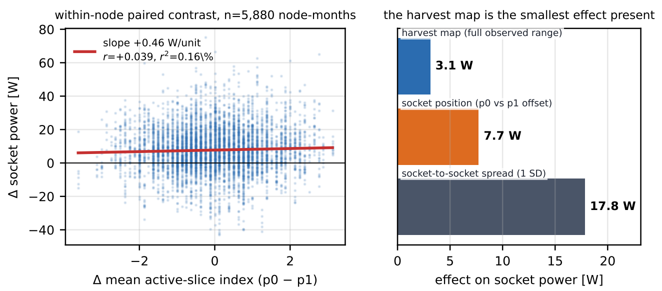

Fig. 12. Operational impact of the harvest map. Left: within-node paired contrast. Right: the harvest map's full observed range moves socket power less than half as much as mere socket position, and a sixth as much as the socket-to-socket spread.

occurs. Maintenance logs would still confirm the cause directly.

4) **[OI-4] Partly resolved.** All references have been verified against their original sources; two were corrected in the process (the negative-binomial model is Koren, Koren and Stapper in *IEEE T. Computers*, not Stapper alone in *T. Semiconductor Manufacturing*). What remains outstanding is the **novelty check**: our claim is an awareness claim, resting on a systematic but necessarily incomplete search.
5) **[OI-3] Defect-driven versus policy-driven harvesting** cannot be separated from one machine; both are reported.
6) **[OI-5] Resolved.** Monthly coverage ranges from 18 % to 99.4 %, but the coverage filter does not drive any result: clean-day rates are near-identical between lots, coverage is uncorrelated with the harvest pattern ($r = +0.004$, $p = 0.88$), and restricting to $\geq$90 % coverage leaves the lot contrast unchanged (Sec. IV-G).
7) **[OI-6, OI-7] both resolved.** The sub-16 socket-days were dominated by an extraction defect on our side, now gated; the nine that survive are the guard episodes of Sec. IV-H. The two nodes appearing to change within a day are same-day reboots.
8) **[OI-8] The mechanism behind p0/p1 relabelling is unknown**; any per-socket longitudinal study on this dataset risks silently mixing two dies.
9) **Common-mode artefact** in the thermal analysis, addressed by same-distance contrasts.
10) **Preprocessing moved our numbers more often than the data did.** We record this because it is the most transferable thing we learned. Three reported effect sizes changed materially during this work, none because of new measurements: a three-day sample overstated the within/cross-slice contrast threefold *and inverted the floorplan conclusion*; a one-minute alignment grid—adopted to work around an unrelated bug and never revisited—inflated that contrast by 2.2×; and an intermediate revision quoted the corrected contrast from a different subsample than the one it named. **Two of the three were introduced while fixing something else.** The defences that worked were mechanical rather than clever: re-deriving every number from a single stated pipeline, varying each discretionary choice (Sec. IV-K), and reporting the most conservative variant rather than the most favourable. Anyone mining a telemetry archive of this kind should expect the same failure mode, and should be at least as suspicious of their corrections as of their original analysis.
11) **[OI-10] Single system**; findings may reflect CINECA procurement rather than POWER9 generally.

VIII. Conclusion

Out-of-band telemetry that supercomputer operators already collect inadvertently publishes a manufacturing secret: which units a vendor fused off each individual die. Recovering these maps for 1,962 POWER9 sockets shows harvesting operates exactly at the slice—the unit sharing an L2/L3 block—with no exceptions in 1.67 M socket-days; that the maps are per-die rather than per-SKU, with 443 of 495 possible patterns realised; that harvested slices cluster in sensor-index space; and that a sharp changepoint at node 439 separates two procurement populations with different binning statistics. Over 2.5 years the maps are static to within 0.25 percentage points per month, change only across reboots, and exhibit three field core-guard episodes, each removing exactly one slice while every other sensor on the socket keeps reporting.

Two things deserve emphasis. The thermal probe recovers the physical slice ordering from telemetry alone, which is what licenses reading the clustering spatially and connects the measurement to the yield literature; it also shows how much this depends on sample size, since a three-day sample gave the opposite answer. And the harvest map turns out to have no measurable operational effect—a negative result we consider as useful as the positive ones, since it tells operators what they can safely ignore. The broader point is that fleet telemetry is a viable instrument for hardware provenance, and that its unintended disclosures deserve attention from those who publish such datasets.

Artefact availability

All analysis code, the derived per-(node, socket, core, day) tables, and the recovered harvest maps are available at the artefact repository accompanying this paper. Every result is reproducible from the public M100 ExaData Zenodo releases [1]; the derived tables (15 MB) suffice for all structural and longitudinal results without re-downloading the 49.9 TB source.

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